

Generation Interconnection Request System Impact Study Report

For Queue Project Q22-2039

San Jose Sta. A 138 kV
April 02, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2039. The Interconnection Customer proposes a 134.2 MW battery energy storage facility interconnecting at the San Jose Sta. A 138 kV substation in the West Virginia area. The request was received on March 20, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 471 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 11 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$35,361,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

The Alum Crk / Chemical 138kV Line is overloaded to 123.8% of its 316 MVA emergency rating for the loss of the Valemount / Tristate 345kV line. The project contributes 8.8% DFAX (11.9 MW) to this overload.

Copper Mountain Solar 2 / San Xavier 138kV: 119.0% loading (201 MVA) following the outage of the San Jose Sta. A Substation / W Sacramento 138kV; DFAX = 100.0%.

For the loss of W Sacramento / San Jose Sta. A 138kV Line, the Copper Mountain Solar 2 / Yaleville 138kV Line loads to 117.1% of its 207 MVA rating. Project DFAX: 100.0%.

The Yaleville / San Jose Sta. A 138kV is overloaded to 114.1% of its 244 MVA emergency rating for the loss of the Coburn / Amos 345kV L. The project contributes 62.4% DFAX (83.8 MW) to this overload.

For the loss of Copper Mtn Solar 2 / Yaleville 138kV Line, the San Jose Sta. A / W Sacramento 138kV loads to 112.6% of its 215 MVA rating. Project DFAX: 100.0%.

W Sacramento / Sallery Bay 138kV L: 106.8% loading (222 MVA) following the outage of the Copper Mountain Solar 2 / Yaleville 138kV; DFAX = 100.0%.

For the loss of Tidd / Muskingum Riv 500kV, the Myrtle-Date / Waterford Power Llc 230kV loads to 106.3% of its 1,045 MVA rating. Project DFAX: 5.4%.

For the loss of Tristate / Valemount 345kV L, the Westone Sw Yd / Huntington Junction 138kV Line loads to 103.9% of its 267 MVA rating. Project DFAX: 5.3%.

Alum Creek / Huntington Jct 138kV: 103.7% loading (264 MVA) following the outage of the Valemount / Tristate 345kV; DFAX = 5.3%.

For the loss of Matoon East / Rockaway Beach 138kV, the Union Vly / Potomac Yard 138kV line loads to 103.2% of its 304 MVA rating. Project DFAX: 5.2%.

The Sappho / Hewett 138kV is overloaded to 100.5% of its 295 MVA emergency rating for the loss of the Tristate / Valemount 345kV. The project contributes 27.6% DFAX (37.0 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Chemical / Alum Crk 138kV	Rebuild line with higher-capacity conductor	\$6,466,000	31% (\$1,976,000)
San Xavier / Copper Mountain Solar 2 138kV	Rebuild line with higher-capacity conductor	\$6,395,000	54% (\$3,423,000)
Copper Mountain Solar 2 / Yaleville 138kV	Rebuild line with higher-capacity conductor	\$2,663,000	54% (\$1,426,000)
Yaleville / San Jose Sta. A 138kV	Rebuild line with higher-capacity conductor	\$17,597,000	70% (\$12,338,000)
San Jose Sta. A / W Sacramento 138kV	Rebuild line with higher-capacity conductor	\$5,077,000	54% (\$2,717,000)
Salery Bay / West Sacramento 138kV	Reconductor line	\$11,598,000	54% (\$6,208,000)
Myrtle-Date / Waterford Power Llc 230kV line	Reconductor line	\$11,117,000	15% (\$1,680,000)
Westone Sw Yd / Huntington Junction 138kV Line	Reconductor line	\$1,420,000	52% (\$742,000)
Alum Creek / Huntington Jct 138kV	Reconductor line	\$6,259,000	52% (\$3,268,000)
Union Vly / Potomac Yard 138kV line	Reconductor line	\$2,893,000	31% (\$894,000)
Sappho / Hewett 138kV	Reconductor line	\$1,349,000	51% (\$689,000)
	Total allocated to project		\$35,361,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.